CMPS 261 Progress Report: NASA Kepler Exoplanet Candidate Classification

Group 4: Joe Habre and Joy Habib Najem — **Track A:** 4.1 Tabular (NASA KOI)
GitHub Repository: `https://github.com/Joeehabre/CMPS-261`

1. Problem Statement

The NASA Kepler Objects of Interest (KOI) cumulative dataset contains $m = 9,564$ transit events observed by the Kepler Space Telescope across 153 archive attributes describing orbital parameters, transit signal geometry, photocenters, and host-star properties. Each observation $x^{(i)}$ is assigned an archive disposition (`koi_disposition`): `CONFIRMED` (2,748, 28.73%), `CANDIDATE` (1,977, 20.67%), or `FALSE POSITIVE` (4,839, 50.60%). Following our course learning framework—defining a hypothesis set $\mathcal{H}$ and learning algorithm $\mathcal{A}$ to approximate the target function $f : \mathcal{X} \to \mathcal{Y}$—our objective is to build a classifier that distinguishes promising exoplanet candidates from astrophysical and instrumental false positives (such as eclipsing binary stars or background stellar contamination). We formulate the primary task as binary classification: **Potential Exoplanet** (`CONFIRMED` ∪ `CANDIDATE`, $y^{(i)} = 1$, 4,725 instances, 49.40%) versus **False Positive** (`FALSE POSITIVE`, $y^{(i)} = 0$, 4,839 instances, 50.60%). Because unconfirmed candidates may contain residual label noise, we also define a secondary strict benchmark restricted to verified dispositions (`CONFIRMED` vs. `FALSE POSITIVE`, $m = 7,587$) to evaluate label-noise sensitivity.

Changes Since Initial Proposal: While our core problem formulation remains unchanged from the initial proposal, we introduced two methodological refinements during baseline development: (1) an explicit pre-vetting feature audit that removes post-classification disposition scores and vetting flags (`koi_fpflag_*`), and (2) **host-star grouped partitioning** (`GroupShuffleSplit` and `GroupKFold` on star ID `kepid`) so multi-candidate stellar systems never straddle training and validation splits.

2. Data Preprocessing

1. **Target Leakage Audit & Feature Filtering:** Several attributes in the cumulative KOI archive are produced *after* human or pipeline classification and leak the target label. Using `isnull().sum()` and archive documentation, we audited all columns and removed:
   - *Post-Vetting Leakage Columns (7):* `koi_disposition`, `koi_pdisposition` (preliminary pipeline disposition), `koi_score` (vetting confidence score in $[0, 1]$), and the four false-positive flags: `koi_fpflag_nt` (not transit-like), `koi_fpflag_ss` (stellar eclipse), `koi_fpflag_co` (centroid offset), and `koi_fpflag_ec` (ephemeris match contamination).
   - *Identifiers & Categorical Provenance Metadata (17):* `rowid`, `kepid` (retained solely as a grouping key), `kepoi_name`, `kepler_name` (assigned only to confirmed planets, directly leaking $y = 1$), `koi_tce_delivname`, and 12 archive provenance/comment strings.
   - *High-Missingness Columns ($\geq$ 50% NaN):* Dropped sparsely populated columns (such as `koi_teq_err1`, `koi_teq_err2`, and `koi_sage`, which are $> 90\%$ to 100% missing), retaining 103 pre-vetting numerical attributes.
2. **Domain Feature Engineering & Skewness Transformation:** To help linear and neural models capture nonlinear physical laws without high-degree polynomial explosion, we engineered 8 domain-specific features ($n = 111$ total features): (i) *Geometric Transit Consistency* comparing observed transit depth δ (`koi_depth`) against the expected planet-to-star area ratio $(R_p/R_*)^2$, as grazing eclipsing binaries violate $\delta \approx (R_p/R_*)^2$; (ii) *Transit Duty Cycle* (`koi_duration/koi_period`); and (iii) $\log(1+x)$ transformations on six heavily right-skewed positive variables (`koi_period`, `koi_depth`, `koi_prad`, `koi_insol`, `koi_model_snr`, `koi_duration`).
3. **Grouped Splitting, Imputation, and Standardization:** Because 9,564 KOIs orbit 8,214 unique host stars (`kepid`), random row splitting leaks stellar properties (`koi_steff`, `koi_srad`) between train and test sets. We partitioned the dataset into an 80% development set ($m_{\text{dev}} = 7,662$ KOIs) and a 20% held-out test set ($m_{\text{test}} = 1,902$ KOIs) grouped by `kepid` (0 shared stars), using 5-fold `GroupKFold` cross-validation on the development set. Missing entries (4.36% overall, including 363 rows missing `koi_srad`) are imputed inside each fold via **Median Imputation** plus binary missingness indicators (`add_indicator=True`), followed by **Z-score Standardization** ($z = \frac{x-\mu}{\sigma}$) for linear and neural models to ensure even gradient descent convergence and fair L_2 regularization. For XGBoost, we leverage native missing-value routing, which learns default split directions that maximize split gain.

3. Evaluation Metrics

- **ROC-AUC and PR-AUC (Ranking Performance):** Because follow-up radial-velocity spectroscopy is resource-constrained, astronomers rank candidates by predicted probability. ROC-AUC evaluates global discrimination across all thresholds via True Positive Rate (TPR = $\frac{TP}{TP+FN}$) vs. False Positive Rate (FPR = $\frac{FP}{FP+TN}$), while Precision-Recall AUC (PR-AUC) directly measures positive-class precision trade-offs across recall targets.
- **F1-Score, Precision, and Recall (Thresholded Classification):** At decision threshold $\tau = 0.5$, False Negatives discard real exoplanets whereas False Positives waste telescope time on stellar binaries. Reporting Precision = $\frac{TP}{TP+FP}$, Recall = $\frac{TP}{TP+FN}$, and their harmonic mean (F1 = $\frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$) quantifies both error types.
- **Expected Calibration Error (Confidence Trustworthiness):** Because valid probabilities in $(0, 1)$ are not automatically calibrated, we measure 10-bin Expected Calibration Error (ECE = $\sum_{b=1}^{10} \frac{|B_b|}{m} |\text{acc}(B_b) - \text{conf}(B_b)|$) and plot reliability diagrams to verify whether predicted confidence matches empirical accuracy.

4. Classical Baseline (Level 1)

We implemented and tuned a structured Level 1 baseline ladder evaluated via 5-fold `GroupKFold` cross-validation:

1. **Linear Baseline (Regularized Logistic Regression):** Linear logit $z = w \cdot x + b$ with Sigmoid activation $f_{w,b}(x) = g(z) = \frac{1}{1+e^{-z}}$, trained under Binary Cross-Entropy loss with L_2 regularization tuned over $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ ($C^* = 0.1$).
2. **Bagging Baseline (Random Forest):** An ensemble of $B = 500$ decision trees trained via bootstrap sampling with replacement and random feature subsets $K = \sqrt{n}$ at each split (`max_depth=16`) to decorrelate trees and reduce variance.
3. **Boosted Baseline (Tuned XGBoost):** Sequential gradient-boosted decision trees trained on first-order gradients (g_i) and second-order Hessians (h_i) of the log-loss objective, optimizing regularized split gain $\text{Gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right] - \gamma$. Hyperparameters were tuned over learning rate $\eta \in \{0.01, 0.03, 0.05, 0.1\}$, `max_depth` $\in \{4, 6, 8\}$, L2 leaf regularization $\lambda \in \{1, 5, 10\}$, split complexity penalty $\gamma \in \{0, 0.5, 1.0\}$, row subsampling (0.8), and column subsampling (0.8), selecting $\eta = 0.03, d = 6, \lambda = 5.0, \gamma = 0.5$.

Model (5-Fold Host-Star Grouped CV)	ROC-AUC	PR-AUC	F1-Score	Precision	Recall	ECE (↓)	Fit Time/Fold
Logistic Regression ($L_2, C = 0.1$)	0.9512 ± 0.004	0.9496 ± 0.011	0.8855 ± 0.009	0.871	0.900	0.0287	0.75 s
Random Forest (500 trees, `max_depth=16`)	0.9683 ± 0.003	0.9704 ± 0.004	$\mathbf{0.9007 \pm 0.008}$	**0.915**	0.884	0.0439	8.98 s
XGBoost (Tuned Classical Baseline)	$\mathbf{0.9709 \pm 0.003}$	$\mathbf{0.9722 \pm 0.004}$	0.8989 ± 0.007	0.906	**0.892**	0.0235	2.40 s
ResNet-MLP (Temp. Scaled, Level 2)	0.9656 ± 0.004	0.9672 ± 0.005	0.8911 ± 0.008	0.889	0.893	**0.0084**	16.50 s

5. Sophisticated Model (Level 2)

- **Architecture & Justification:** For Level 2, we implemented a deep tabular **Residual Multi-Layer Perceptron (ResNet-MLP)** with residual skip connections ($y = F(x) + x$), Batch Normalization, ReLU hidden activations, Dropout ($p = 0.25$), and Sigmoid output activation, trained with mini-batch AdamW ($\alpha = 10^{-3}$, weight decay 10^{-4}, batch size 128) and Early Stopping. While decision trees partition continuous orbital laws ($P^2 \propto a^3/M_*$) via axis-aligned step functions, a residual neural network learns smooth hierarchical representations across coupled orbital and stellar features while skip connections preserve gradient flow. To ensure trustworthy confidence estimates, we freeze trained weights and apply post-hoc **Temperature Scaling** ($\hat{p} = \sigma(z/T)$), optimizing scalar T on a held-out validation subset.
- **Current Progress & Preliminary Results:** Our ResNet-MLP substantially outperforms Logistic Regression (+0.0144 ROC-AUC, +0.0176 PR-AUC), confirming effective non-linear representation learning. Crucially, post-hoc Temperature Scaling achieves the best probability calibration across all models (ECE = 0.0084, compared to 0.0235 for XGBoost and 0.0439 for Random Forest) without altering classification ranking or accuracy. In raw ranking metrics, standalone ResNet-MLP slightly trails tuned XGBoost (0.9656 vs. 0.9709 ROC-AUC) while requiring $\sim 6.9\times$ longer training time, reflecting the strong inductive bias of gradient-boosted trees on tabular thresholds. Next, we will evaluate a calibrated **XGBoost + ResNet-MLP ensemble** and a **Tabular Self-Attention (FT-Transformer)** model.

6. Remaining Plan and Risks (Until November 20)

- **Remaining Experiment Schedule:**
  - *Nov 6–Nov 10 (Level 2 Tuning & Ensembling):* Tune FT-Transformer and ResNet-MLP architectures (depth, cosine learning-rate schedule) and evaluate probability-calibrated tree + neural ensembles on the held-out test set.
  - *Nov 11–Nov 14 (Split Sensitivity & Feature Ablation):* Quantify sensitivity to evaluation design by comparing (i) Host-Star Grouped Split vs. Random Split (measuring host-star leakage), (ii) Candidate grouping vs. Strict Confirmed-only disposition, and (iii) an ablation isolating core transit/stellar physical features (36 columns) from centroid/secondary-eclipse diagnostic statistics (`koi_dicco_*`, `koi_fwm_*`) and post-hoc flags (`koi_fpflag_*`).
  - *Nov 15–Nov 20 (Failure Mitigations, Complexity Trade-offs & Final Deliverables):* Benchmark parameter counts, training/inference latency, and performance-vs.-complexity Pareto curves; finalize the poster, LaTeX report, and GitHub model weights.
- **Early Observations on Model Failures:** Stratified error analysis of our validation predictions highlights four systematic failure regimes: (1) **Weak Signals (`koi_model_snr` < 12):** error rates more than double compared to high-SNR transits (SNR > 50) as shallow planetary transits merge with photometric noise; (2) **Grazing Eclipsing Binaries (`koi_impact` > 0.9, $R_p \in [8, 22]\, R_\oplus$):** V-shaped stellar eclipses closely mimic gas-giant planets; (3) **Evolved Giant Stars (`koi_slogg` < 3.8, $R_* > 2.5\, R_\odot$):** diluted transit depths reduce recall compared to solar-type dwarfs; and (4) **Missing Stellar Parameters (363 rows):** objects missing `koi_srad` suffer elevated misclassification due to missing radius ratios.
- **Risks & Mitigations:** Deep tabular networks risk overfitting noisy features and trailing gradient-boosted trees in raw AUC. Even if standalone deep models do not surpass XGBoost in AUC, demonstrating superior confidence calibration (ECE = 0.0084), ensemble complementarity, and computational trade-offs directly answers the project's complexity question.

Appendix: Preliminary Experimental Figures (Excluded from 2-Page Text Limit)

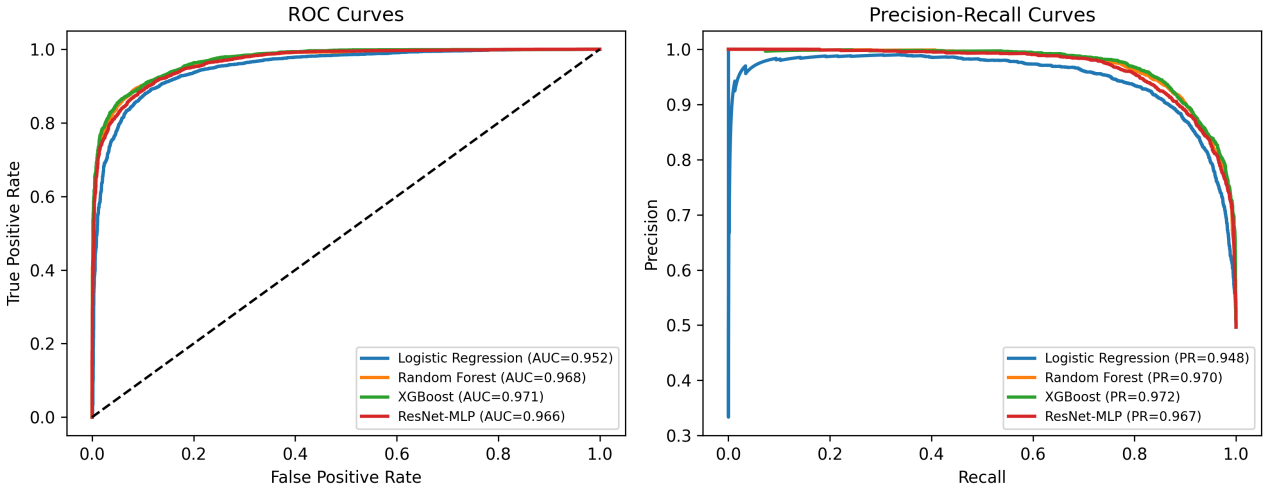


Figure 1: 5-Fold Host-Star Grouped Cross-Validation ROC (left) and Precision-Recall (right) curves comparing Level 1 classical baselines (Logistic Regression: blue, Random Forest: orange, Tuned XGBoost: green) against the Level 2 ResNet-MLP (red).

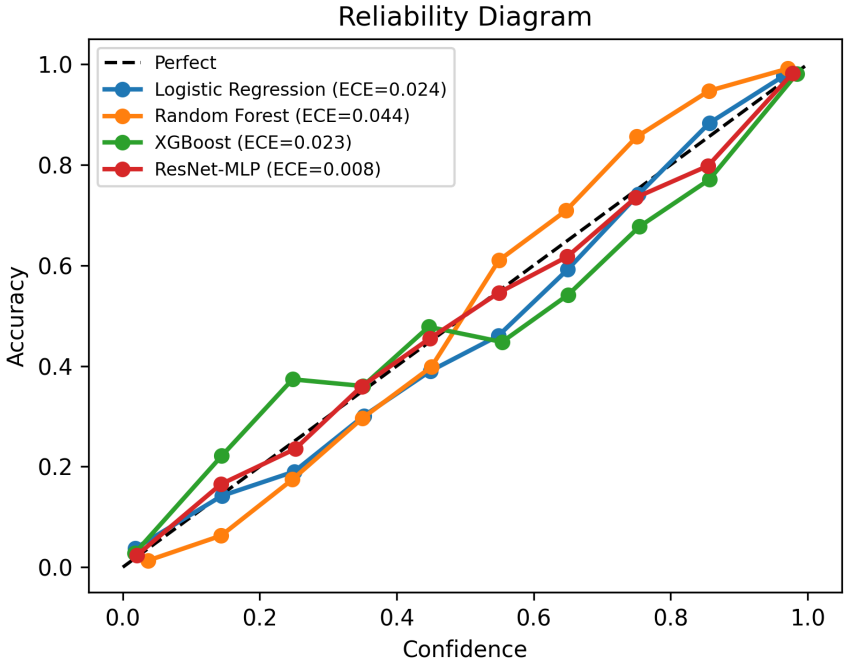


Figure 2: Reliability diagram demonstrating confidence calibration across 10 probability bins on 5-fold grouped cross-validation. Post-hoc Temperature Scaling on frozen ResNet-MLP logits achieves near-perfect calibration (ECE = 0.008), outperforming uncalibrated tree ensembles (ECE = 0.023–0.044) without changing classification decisions.